ARPITA MALLIK

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Chittagong - Bangladesh

EXPERIENCE

- **ASSRO, CUET** *2025 – 2026*

  Machine Learning Instructor Chittagong, Bangladesh
  - Mentored junior students in Machine Learning, Deep Learning, NLP, and Large Language Models through technical sessions and hands-on workshops.
  - Guided research and implementation for NLP shared-task projects, contributing to workshop paper publications in ACL Anthology venues.

- **Green Lead** *June 2024 – December 2025*

  Tech Solutions Architect Remote
  - Maintained and updated the Green Lead website, ensuring optimal performance and user experience.
  - Assisted in troubleshooting and resolving technical issues related to website functionality and maintenance.

EDUCATION

- **Chittagong University of Engineering & Technology** *March 2022 - July 2026*

  BSc. in Computer Science & Engineering
  - GPA: 3.62/4.00

SKILLS

- **Programming Languages:** Python, SQL
- **Machine Learning:** PyTorch, scikit-learn, Neural Networks, Model Evaluation
- **NLP & LLMs:** Natural Language Processing (NLP), Transformers, Hugging Face, LangChain, Prompt Engineering, LoRA Fine-Tuning
- **RAG & Vector Databases:** Retrieval-Augmented Generation (RAG), Embeddings, Pinecone, Semantic Search
- **Backend & Deployment:** FastAPI, Flask, Docker, AWS EC2, AWS S3, GitHub Actions
- **MLOps & Tools:** MLflow, DVC, Git, GitHub, CI/CD

PROJECTS

- **Vehicle Insurance Prediction – End-to-End MLOps System**

  Tools: Python, scikit-learn, MongoDB Atlas, AWS (S3, ECR, EC2, IAM), Docker, GitHub Actions, Flask
  - Built a modular end-to-end ML pipeline for vehicle insurance prediction.
  - Implemented schema-based validation and artifact-driven model training and evaluation.
  - Designed a versioned S3 model registry with threshold-based promotion.
  - Deployed a Dockerized Flask service on AWS EC2 with CI/CD via GitHub Actions.

- **Stress Monitoring Pipeline — Real-Time Load Testing and Metrics**

  Tools: Python, FastAPI, Prometheus, Grafana, k6, Docker
  - Built an end-to-end load testing and real-time monitoring pipeline using FastAPI, Prometheus, and Grafana with configurable k6 load tests.
  - Instrumented FastAPI application with Prometheus metrics (request counters, latency histograms) and designed Grafana dashboards for visualizing performance metrics including request rate, P95/P99 latency, and app availability.
  - Containerized with Docker for seamless deployment and configured automated metric scraping every 3 seconds.

- **Medical AI Chatbot – Memory-Based RAG Medical QA System**

  Tools: Python, Flask, LangChain, Pinecone, HuggingFace Embeddings
  - Developed a medical QA chatbot using a memory-augmented RAG pipeline with semantic retrieval via Pinecone.

- Implemented Flask backend with LangChain for chunking, retrieval, and session-aware conversational history.

- **Container Job Orchestrator — Real-Time Job Execution Platform**
  Tools: Python, FastAPI, Redis, PostgreSQL, Docker, WebSockets
  - Built a full-stack job execution platform with Redis queue-based job dispatch and Docker container execution managed by distributed workers.
  - Implemented real-time log streaming via WebSockets and Redis Pub/Sub, with PostgreSQL persistence and asynchronous status tracking.
- **End-to-End Boston House Price Prediction**
  Tools: Python, Flask, Docker, Git, GitHub
  - Built a full-stack ML web app to predict house prices using features like crime rate and tax.
  - Dockerized the app for seamless deployment and added REST API support with Postman.
- **CUET Thesis & Project Portal**
  Tools: React JS, Firebase
  - Built a centralized portal for managing and browsing CUET students' thesis and projects.
  - Used Firebase for real-time storage, retrieval, and user authentication.

Publications

C=Conference, J=Journal,S=In Submission, T=Thesis

[C.1] **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language**. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*
DOI: 10.18653/v1/2025.dravidianlangtech-1.57

[C.2] **CUET-823 at SemEval-2026 Task 9: LoRA-Based Instruction-Fine-Tuned LLMs vs. Transformer Models for Bengali Polarization Detection**. Arpita Mallik, Ratnajit Dhar. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

Honors and Awards

Finalist – DL Sprint 4.0 *February 2026*
BUET CSE Fest 2026
- Ranked among the Top 18 teams out of 192 in a nationwide deep learning competition.
- Built a pipeline for Bangla long-form speech recognition and speaker diarization using Whisper-based ASR and fine-tuned diarization models.

Champion – NodeScape *July 2025*
CUET Computer Club
- Secured 1st place in CUET Computer Club's flagship 3-phase hackathon, NodeScape.
- Developed an AI-powered graph classification system with algorithm visualization and Dockerized CI/CD deployment using React, Flask, and Python.

Finalist – Intra CUET Machine Learning Contest *July 2025*
IEEE CS CUET Student Branch Chapter
- Selected as a finalist in intra-university ML contest.
- Built a Bangla physics MCQ question-answering system leveraging large language models (LLMs) to understand and solve concept-based queries.

Top 7 Finalist – HackCSB 2024 *August 2024*
Coding Shikhbe Bangladesh
- Achieved finalist position out of 116 teams at HackCSB for the project "MotivationMate."
- Demonstrated creativity and teamwork in building a mental health-focused web application.

Certifications

- **Google Cloud Fundamentals: Core Infrastructure** - Coursera *June 2025*
- **Introduction to Shell** - DataCamp *Dec 2025*
- **AWS Concepts** - DataCamp *Nov 2025*
- **Career Essentials in Generative AI** - Microsoft & LinkedIn *July 2024*
- **Supervised Machine Learning: Regression and Classification** - Coursera *February 2024*
- **Advanced Learning Algorithms** - Coursera

Leadership Experience

- **Vice Chair (Machine Learning Wing)** *July 2025 – Present*
  IEEE CS CUET Student Branch Chapter